Fiscal Year:	FY 2014	**Task Last Updated:**	FY 09/25/2014
PI Name:	Thiessen, David Brian Ph.D.		
Project Title:	Phase Separation and Phase Change Using Helically-Supported Capillary Channels		
Division Name:	Physical Sciences		
Program/Discipline:	FLUID PHYSICS		
Program/Discipline--Element/Subdiscipline:	FLUID PHYSICS--Fluid physics		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	99164-2814	**Congressional District:**	5
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2008 Fluid Physics NNH08ZTT002N
Start Date:	06/01/2009	**End Date:**	05/15/2014
No. of Post Docs:		**No. of PhD Degrees:**	2
No. of PhD Candidates:	3	**No. of Master' Degrees:**	1
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	1
No. of Bachelor's Candidates:	3	**Monitoring Center:**	NASA GRC
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Flight Program:			
Flight Assignment:	NOTE: Extended to 5/15/2014 per NSSC information (Ed., 4/5/2013) NOTE: New end date is 5/31/2013 per NSSC (Ed., 4/18/2012)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Marston, Philip (Washington State University)		
Grant/Contract No.:	NNX09AI82G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Wicking structures with low resistance to axial flow are currently used in heat pipes and may have other applications in microgravity associated with phase separation and phase change technologies. The proposed work involves a fundamental study of a novel capillary flow channel consisting of a helical-wire support that is potentially useful in technologies and benefits from having a symmetry that simplifies the analysis of stability and flow dynamics. The channel geometry makes it particularly easy to measure the capillary dynamics on a global scale using high-speed imaging and image analysis. It is proposed to study the capture of droplets from a flowing air stream by a liquid-filled channel and the capture of bubbles from a flowing liquid stream by a gas-filled channel. Phenomena to be understood include the drop or bubble coalescence event and the subsequent response of the channel as capillary waves propagate from the impact zone dissipating energy. A model for capillary wave dynamics and for dynamic flows in these channels will be developed. Models for drop and bubble capture will also be developed that will be coupled to the channel dynamics model. Studies of phase change at an extended meniscus supported by the helical wire are also proposed. In all cases the basic component to be studied is the capillary channel supported by a helical wire and connected to a constant-pressure reservoir. The behavior of multichannel arrays for phase separation will also be investigated. Parabolic flight experiments are planned and ISS facilities are identified for possible future experiments. The understanding gained from this project is expected to be applicable to other channels that are based on the same fundamental physics. The experimentally validated physics-based models developed from this work might help guide the design of heat pipes and of novel phase-separation devices.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The experimentally validated physics-based models developed from this work might help guide the design of heat pipes and of novel phase-separation devices. The channels being developed have potential as variable-pitch superhydrophobic channels with small resistance to flow.
	An important goal of the project is to establish the performance limits of capillary channels that use helical support structures so that decisions can be made as to the suitability of such structures for applications in space such as cryogen control in tanks, phase separation in Rankine-cycle power systems, or in various life-support processes. Physics-based models were developed for channel dynamics including transient behavior and steady flow properties. These models will be helpful in designing phase separators for particular applications. Capillary flow limits as large as 628 mL/min for a single large-bore channel have been measured indicating that a phase separator based on such channels would have a very useful capacity. The integration of capillary channels into a phase separation system has been tested on low-gravity aircraft flights and shows significant promise. Further testing under low-gravity conditions using a drop tower are refining issues found with system integration during the tests on the low-gravity aircraft. One of the primary objectives of the project was to elucidate the dynamics of drop capture by a channel with high-speed video. In the first year of the project, an experimental apparatus was constructed to study the capture of falling drops by a single channel that runs between two transparent manifolds, each connected to the same constant-pressure reservoir. Transparent manifolds allow for end-on viewing of the drop collision and absorption process. Drops were allowed to fall onto the channel from a needle that could be positioned at different heights to vary the impact speed and at different lateral positions to vary the so-called impact parameter (lateral distance of drop center from channel center). High-speed imaging of the drop absorption process was performed by viewing through the manifolds using a telecentric lens and collimated back lighting. Channels of 3-mm diameter were found to absorb 90% of the mass of 3-mm drops impacting at 0.6 m/s over a range of impact parameters (0-1 mm). The ability of a primed water-filled channel to continuously absorb water from a two-phase flow was demonstrated under normal gravity using a fine mist generated by an airbrush and directed horizontally at a right angle to the channel. The speed of the entraining air was 7.5 m/s and it was estimated based on rough measurements that the channel was continuously absorbing approximately 50% of the mist droplets incident on it. Although this is a very rough measurement it is not inconsistent with what would be predicted from the inertial impaction collision efficiency. Another objective of the project was to develop a model for capillary wave dynamics and for dynamic flows in the capillary channels. Channel diameter oscillations detected shortly after drop impact are believed to represent a channel resonance indicating that capillary waves are traveling back and forth the length of the channel several times before this energy is dissipated. This effectively disperses drop impact energy over the entire channel. Wave packets are seen to leave the impact region within a few milliseconds thus reducing the pressure in the impact zone and allowing the attached drop to be sucked into the channel. This capillary wave action is seen to die out within a few hundred milliseconds. Because of its apparent importance in the drop-capture mechanism for large drops, capillary waves on a periodically-supported liquid cylinder have been studied theoretically by a physics PhD student (Likun Zhang, PhD May 2012). Efficient transmission of wave energy is seen to occur under conditions of resonance of the individual liquid bridges between adjacent support rings. One of the mechanisms of wave energy dissipation, namely that associated with contact line motion, has also been studied. The theory considers infinitesimal wire support rings that constrain the contact line. Because solid-liquid contact is infinitesimal viscous boundary layer effects are neglected. Potential flow is assumed in the bulk with appropriate kinematic and dynamic boundary conditions on the free surface and a dynamic contact line condition. Capillary waves on a liquid cylinder supported by a spring can be approximated by waves on a liquid cylinder supported by a periodic array of rings. The first step in analyzing such waves is to examine the scattering of axisymmetric capillary waves on a liquid cylinder by a concentric infinitesimal ring. Time-harmonic capillary waves on a liquid cylinder in zero gravity axially incident on and scattered by an infinitesimal concentric barrier were considered theoretically in the inviscid limit. This study investigates the importance of contact line dynamics on the control of liquids in low gravity. Edge conditions on the barrier that are investigated include a pinned contact line and a moving contact line with no hysteresis by Hocking's effective-slip model. The effective-slip boundary condition assumes that contact line velocity is proportional to the deviation of the contact angle from equilibrium multiplied by a slip coefficient. A matched evanescent-wave eigenfunction expansion was used to satisfy continuity and edge conditions at the barrier. Scattering and dissipation by the contact line were determined as a function of wavenumber and slip coefficient. Contact line dissipation at the barrier was found to have a maximum for incident waves whose phase velocity is on the order of the slip coefficient. The short-wave limit agrees with established results for the scattering of planar gravity-capillary waves on a deep liquid by a surface-piercing vertical barrier in the limit of zero barrier depth and zero gravity. The results were published in 2013 (J. Fluid Mech., vol. 719, pp. 295-313, (2013)).

Task Progress:

A theoretical model was also developed for the propagation of capillary waves on a simplified axisymmetric channel, that is a liquid cylinder supported by an equally spaced array of concentric rings. The theory is based on the Bloch theorem and an eigenfunction expansion. Wave speeds and their dependence on channel parameters were computed from the semi-analytical solution. The phase and group velocities in the first pass band were calculated and it was found that both the phase and group velocities decrease with the Bloch wavenumber. The group velocity was found to be generally less than the phase velocity and only in the long wave limit do they approach the same maximum velocity.

An automated experiment developed for drop-capture measurements was also used for measuring the speed of wave packets that emanate from the impact zone in a separate set of experiments with a camera axis orthogonal to the channel. An image analysis code developed during year 2 measures the position of the meniscus between adjacent coils of the spring relative to a reference image taken just prior to drop impact. Using a laser drop detector as a time reference and the fact that the waves emanating from the impact zone are reproducible, it has been possible to make a panoramic movie of wave propagation over a large portion of the channel by combining frames of high-speed videos of adjacent sections of the channel. The speed of the leading wave packet seen in the panorama was measured for channels with different pitches and compared with the infinitesimal-wire theory. A somewhat higher speed for the measured wave-packets can be understood as the effect of finite wire size since this leads to shorter sections of free surface and a stiffer channel.

Flow dynamics are important for understanding the phase-separation capacity of a multichannel device. It is helpful to know the flow resistance and maximum flow capacity of a channel as a function of channel diameter, pitch, and aspect ratio. Experiments of this type in low-gravity conditions would require a long-duration low gravity platform. To make measurements under normal gravity for large-bore channels requires coating the springs with a hydrophobic coating to avoid channel failure. Baba Abdul (PhD in Chemical Engineering, May 2014) developed a procedure to uniformly coat a spring with a superhydrophobic coating and then studied both the hydrostatic properties and steady flow dynamics in single channels. The hydrostatic measurements give channel compliance and stable pressure limits. Stable pressure limits are important for predicting capillary flow limits. He also developed a numerical model for both fully-developed flow and for flow in finite-length channels where flow-development effects are important.

A sol-gel coating system to produce superhydrophobic thin-film aerogels was identified in year 2 and a lathe was built for uniformly coating this aerogel film on springs using an airbrush to spray the spring as it is rotated from both ends. Water was flowed in channels with length-to-diameter ratios ranging between 32 and 200 at flow rates up to 628 mL/min. Pressure drop measurements were made at many different flow rates and for springs with different pitch and diameter. The slip length for a smooth tube of the same inside diameter as the spring that achieves the same pressure drop when entrance conditions and length-to-diameter ratio are the same is taken as a measure of the effective slip of the channel wall. Slip lengths in the vicinity of 26 microns were measured for a 1/4-inch spring and slip lengths between 70 and 140 microns for a 1/8-inch spring. These are at the high end of measured slip lengths reported in the literature. The maximum flow rate achieved for a 1/8-inch spring with a dimensionless pitch of 0.4 and aspect ratio of L/D=111 was 93.4 mL/min and for a 1/4-inch spring with a dimensionless pitch of 0.3 and aspect ratio of L/D=65 was 628 mL/min. This difference is roughly what would be expected based on a simple scaling analysis when one takes account of the difference in aspect ratio, pitch, and diameter of the channels. It should be noted that if the single-channel flow rates measured here were duplicated in a multi-channel square array with center to center spacing of twice the channel diameter, the superficial mass flux in common engineering units for 1/8-inch springs would be 28406 lb/hr ft^2 and for the 1/4-inch springs would be 47748 lb/hr ft^2. These rates are in a range typically seen in ground-based absorption towers where the liquid flow is driven by gravity through a packing.

Flow simulation for the helical geometry of a liquid-filled spring is three-dimensional and computationally expensive; however, some aspects of the flow in such a channel can be approximated by flow in the corresponding axisymmetric geometry of an array of wire rings. Flow in such a channel is modeled by the finite element method on a periodic domain. The finite element model was validated by comparison to semi-analytical solutions available in the literature for related geometries. Drag reduction is found to increase with increasing pitch of the structure, which increases the fraction of slip boundary, and to decrease with increasing roughness of the composite wire-plus-meniscus boundary and with Reynolds number above 50. The fraction of slip boundary is increased in proportion to the pitch and also increases with increasing contact angle. Boundary roughness depends on contact angle, wire radius, and the channel pressure that in turn affects the meniscus curvature. Inertial effects become more pronounced above a Reynolds number of 50. The highest drag reduction is found to occur for the highest contact angle, Reynolds numbers below 50 and for a channel pressure that gives a smooth liquid-gas interface (no-shear boundary).

A physics PhD student (Maverick Terrazas) is working on the integration of channels into systems and the testing of those systems under low-gravity conditions. The limited low-gravity time available to us (two days of parabolic flights) has motivated the use a nearby drop tower facility to test certain aspects of our systems. The purpose of the flight experiment was to demonstrate that a multichannel array can separate the liquid from an air flow of 1 m/s in a 4-inch square duct with a volume fraction of water of about one part per thousand. The data acquired included two axes of telecentric video, 3-axis accelerometer, and pressure drop between the channel manifold and the reservoir. The flight experiment tested channels that were too large to be tested under 1-g conditions. The experiment was flown on two consecutive days, May 3 and 4, 2012 with 40 parabolas on each day. A fairly uniform two-phase flow was created in the duct with a velocity that appeared to be of the right order of magnitude. Air bubbles in the manifolds prevented easy viewing from the axial camera, however there were cases where channel filling could be confirmed. A filled channel is confirmed when the interior of the channel shows up as bright white indicating that water is continuous throughout the channel and manifolds. In one case, 10 of 14 channels within the field of view were filled and remained filled during two-phase flow. Because of problems with filling the manifolds not all channels could be filled at once and this prevented lowering the manifold pressures as needed to suck in impacting drops. The array of channels did very efficiently collect impacting drops because no small drops were seen below the array section. It is believed that the liquid from impacting drops collected in the spaces between channels. Because this collected liquid was not pulled into the manifolds it was eventually forced out the bottom of the array by the airflow and could be seen on hand-held video coming out the bottom of the array as very large consolidated masses of liquid.

Experiments were conducted in short duration low-gravity at the Dryden Drop Tower (DDT) facility at Portland State University with the cooperation of Professor Mark Weislogel in order to test a better method of manifold filling than that used in the low-gravity aircraft experiments. Two sets of experiments were conducted: one in March of 2014 and one in June of 2014 with a focus on manifold priming and channel priming, respectively. A new test section was designed with interchangeable single and double spring channels connected between two partially conic manifolds. High-speed video of the manifolds and the channels taken over a two second drop time is still being analyzed. The video

	data will give further insight on the mechanisms of manifold and channel filling for a parameter range that cannot be probed under normal gravity. Furthermore, while the number of channels is significantly reduced from the flight experiment's test section, the drop tower tests cover a much larger range of pitch. Preliminary indications would suggest that it is possible to prime both single and double channel arrays quickly. Another goal of the project was to examine the capture of air bubbles by an air-filled hydrophobic spring submerged in water. A 1/8-inch diameter stainless steel spring coated with a superhydrophobic sol-gel coating was submerged in a water tank and the ends connected by tubing to the open air. A pump was used to push air into a perforated tube at the bottom of the water tank below the air-filled spring such that the bubbles impacted the spring as they floated upward. It was found that nearly 100% of these bubbles were absorbed by the spring for a short time at airflow rates up to 540 mL/min and 183 bubbles/sec. The average bubble diameter was on the order of 4 mm. Bubbles are seen to be absorbed in about 8 ms from the time they touch the spring. Unfortunately the superhydrophobic coating is not robust in this application and is quickly eroded such that bubbles are no longer absorbed after about one second. A commercially available coating is currently being evaluated for this application and shows promise for being much more robust. For a separator of a given volume that is separating liquid from a gas-continuous stream, the liquid volumetric flow rate capacity based strictly on capillary flow limits, scales with the diameter of the channels being used. From this standpoint one would prefer to use large-bore channels. However, larger channels will be more susceptible to failure from the dynamic pressures caused by the gas phase flowing past the channels. Optimal channel diameter would have to be determined for a particular application. Measured capillary flow capacities of horizontal large-bore channels in normal gravity suggest that a multi-channel separator could have a capacity (in terms of superficial mass flux) similar to gravity-driven two-phase contacting towers used in the chemical process industries. Considering the high absorption efficiency found for single channels and the demonstrated ability to prime channels under low-gravity conditions it seems likely that a phase separator consisting of only a few rows of closely-spaced springs (like the one tested on low-gravity flights) could be made to work with further development.
Bibliography Type:	Description: (Last Updated: 09/14/2018)
Dissertations and Theses	Abdul B. (Baba Abdul) "Part I: Assessments of Hands-On Learning and Lecture Part II. Stability and Drag Reduction in Steady Capillary Flow Through Hydrophobic Channels with Minimal Helical Wire Supports." Dissertation, Washington State University, May 2014. , May-2014
Dissertations and Theses	Zhang L. (Likun Zhang) "I. Radiation Forces and Torques of Acoustic Beams II. Capillary Waves in Periodically Supported Channels." Dissertation, Washington State University, May 2012. , May-2012
Dissertations and Theses	Chandurwala F. (Fahim Chandurwala) "Drop Capture by a Helically-Supported Capillary Channel and the Resulting Capillary Waves." Thesis, Washington State University, May 2012. , May-2012